

Day 6 — Near 1000: Padding the Right Part to 3 Digits

Module: Nikhila Multiplication · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will solve 3-digit $\times$ 3-digit multiplications where both factors are within ~ 15 of 1000, correctly padding the right part to 3 digits.

Materials

- Whiteboard.
- Drill sheet (8 problems near 1000).
- Reference card: “deficits from 1000” (a few worked: $998 \rightarrow 2$, $991 \rightarrow 9$, $988 \rightarrow 12$, $1003 \rightarrow +3$, etc.).

Lesson flow

Warm-up (5 min)

- Mixed sign-and-magnitude round: teacher says a number, class says deficit (if below 1000) or excess (if above).
- $1003 \rightarrow +3$. $998 \rightarrow -2$. $991 \rightarrow -9$. $1012 \rightarrow +12$. $988 \rightarrow -12$.

Core (15 min) — Same rule, bigger pad

1. **The rule is unchanged.** Set up the vertical line, find deficits/excesses, cross-subtract or cross-add, multiply the deficits/excesses.
2. **The ONE change: pad the right part to 3 digits.** Base 1000 has 3 zeros. The right part must be exactly 3 digits.
3. **Walk through 998×997 :**
 - Deficits from 1000: 2, 3.
 - LEFT: $998 - 3 = 995$. (Or $997 - 2 = 995$. ✓)
 - RIGHT: $2 \times 3 = 6$. Pad to **3 digits**: 006.
 - Answer: **995006**. ✓ (Calculator: $998 \times 997 = 995006$.)
4. **Walk through 1003×1004 :**
 - Excesses: +3, +4.
 - LEFT: $1003 + 4 = 1007$.
 - RIGHT: $3 \times 4 = 12$. Pad to 3 digits: 012.
 - Answer: **1007012**. ✓
5. **The overflow case: 988×985 :**
 - Deficits: 12, 15.

- LEFT: $988 - 15 = 973$. (Or $985 - 12 = 973$. ✓)
- RIGHT: $12 \times 15 = 180$. Pad to 3 digits: 180 (already 3 digits, no overflow).
- Answer: **973180**. ✓

6. **The harder overflow: 985×978 :**

- Deficits: 15, 22.
- LEFT: $985 - 22 = 963$. (Or $978 - 15 = 963$. ✓)
- RIGHT: $15 \times 22 = 330$. Already 3 digits.
- Answer: **963330**.
- Verify: $985 \times 978 = 963330$. ✓

7. **The hard overflow: 970×965 :**

- Deficits: 30, 35.
- LEFT: $970 - 35 = 935$.
- RIGHT: $30 \times 35 = 1050$. That's 4 digits — overflow! Carry the leading 1: left = $935 + 1 = 936$; right = 050.
- Answer: **936050**. ✓

8. **Mixed-sign case near 1000: 1002×998 :**

- Excess +2, deficit -2.
- LEFT: 1000 (either cross).
- RIGHT: $+2 \times -2 = -4$.
- Answer: $1000 \times 1000 - 4 = 999996$. ✓

Activity (20 min) — Drill

Drill sheet:

- | | | |
|----------------------|-----------------------|-----------------------|
| 1. 998×997 | 2. 996×995 | 3. 999×998 |
| 4. 988×985 | 5. 1003×1004 | 6. 1005×1002 |
| 7. 1002×998 | 8. 970×965 | |

- 10 min individual silent.
- 10 min paired check + redo errors.

Answer key (teacher): | Q | Sign | Deficits/excesses | Left | Right (padded) | Answer | |—|—|—|—|
 —|—|—|—| 1 | below | 2, 3 | 995 | 006 | 995006 | | 2 | below | 4, 5 | 991 | 020 | 991020 | | 3 | below |
 1, 2 | 997 | 002 | 997002 | | 4 | below | 12, 15 | 973 | 180 | 973180 | | 5 | above | 3, 4 | 1007 | 012 |
 1007012 | | 6 | above | 5, 2 | 1007 | 010 | 1007010 | | 7 | mixed | +2, -2 | 1000 | -4 | 999996 | | 8 |
 below | 30, 35 | 935+1=936 | 050 | 936050 |

Wrap-up (5 min)

- **Formative quiz Part C (3 min)** — see quizzes/formative.md.
- Preview Day 7: “*Tomorrow we prove why this works. One line of algebra.*”

Student-facing content

Near 1000 — 3-digit pad

The rule is unchanged. The **only** change: pad the right part to **3 digits** (because base 1000 has 3 zeros).

Examples: - $998 \times 997 \rightarrow 995 \mid \mathbf{006} \rightarrow \mathbf{995006}$ - $1003 \times 1004 \rightarrow 1007 \mid \mathbf{012} \rightarrow \mathbf{1007012}$ - $970 \times 965 \rightarrow 935+1=936 \mid \mathbf{050} \rightarrow \mathbf{936050}$ (right part overflowed) - $1002 \times 998 \rightarrow 1000 \mid -4 \rightarrow \mathbf{999996}$ (mixed)

General rule: the right-part pad-width equals the number of zeros in the base.

Homework

- 8 problems on the back of today's worksheet, mixed near-1000.
- Compute one of them with long multiplication. How much longer did long take?

Differentiation

- **Tier up:** Try 9998×9997 (base 10000). Right part pad = 4 digits. Confirm answer: 99950006.
- **Tier down:** Stay with the easy near-1000 problems (998×997 , 999×998). Build up.

Teacher notes

- The padding rule is the only new thing today. If students have Day 3 fluent, this lesson is mostly drill.
- Forgetting to pad is the most common error. A student who writes $998 \times 997 = 9956$ instead of 995006 has forgotten to pad to 3 digits. The diagnostic: their answer has the wrong number of digits.
- The “general rule” (pad-width = number of zeros in base) is a real generalisation worth boxing on the board. It sets up Day 7's algebra.

Citation(s) used in this lesson

- *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center, Sammamish WA) — Nikhilam multiplication, base-1000 examples.